Schedule

LearnHub:

- Practice problems
- Reading assignments
- Study questions
- Formula sheet
- Class exercises and their solutions

Thursday lecture → PS

Thursday lab → Office hour of TAs

Midterm: Friday, April 25th at 19:00-21:00, on LearnHub, multiple choice, in labs SNAB149 and SNAB220

- Safe Exam Browser, only Excel can be used
- Seating charts will be announced
- LearnHub material will be available.

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Agenda

Short review

Class exercise

Two population tests

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Hypothesis/Significance Tests

HYPOTHESIS

Parameter of interest and its estimator

Null/alternative hypotheses

Test statistic and its distribution (assuming the null hypothesis is true)

SIGNIFICANCE

Parameter of interest and its estimator

Null/alternative hypotheses

Test statistic and its distribution (assuming the null hypothesis is true)

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Test statistics and their distributions (true mean, μ)

- $X_1, \ldots, X_n$: A random sample with E[X] = μ
- We are looking for a random variable $Y=h(X_1,\ldots,X_n; \mu)$ whose distribution can be specified.

	X is normal		X is not normal		
	n<40	n≥40	n<30	30≤n<40	n≥40
σ: known	$\frac{\bar{x}-\mu}{\sigma/\sqrt{n}} \sim Z$	$\frac{\bar{x}-\mu}{\sigma/\sqrt{n}} \sim Z$	?	$\frac{\bar{x}-\mu}{\sigma/\sqrt{n}} \sim Z$	$\frac{\bar{x}-\mu}{\sigma/\sqrt{n}} \sim Z$
σ: unknown	$\frac{\bar{x}-\mu}{s/\sqrt{n}} \sim T_{n-1}$	$\frac{\bar{x}-\mu}{s/\sqrt{n}} \sim Z$	?	? $\frac{\bar{x}-\mu}{s/\sqrt{n}} \sim T_{n-1}$?	$\frac{\bar{x}-\mu}{s/\sqrt{n}} \sim Z$

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Test statistics and their distributions $(\mu_X - \mu_Y)$

$$\sigma = \sqrt{\frac{\sigma_X^2}{n_X} + \frac{\sigma_Y^2}{n_Y}}$$

$$s = \sqrt{\frac{s_X^2}{n_X} + \frac{s_Y^2}{n_Y}}$$

- $\{X_1, ..., X_n\}$, $\{Y_1, ..., Y_m\}$: Two random samples independent of each other with $E[X] = \mu_X$ and $E[Y] = \mu_Y$ with $Var(X) = \sigma_X$ and $Var(Y) = \sigma_Y$
- We are looking for a random variable $U=h(X_1,...,X_n; Y_1, ..., Y_m; \mu_X, \mu_Y)$ whose distribution can be specified.

	X is normal		X is not normal		
	n<40	n≥40	n<30	30≤n<40	n≥40
σ: known	$\frac{\bar{x} - \bar{y} - (\mu_X - \mu_Y)}{\sigma} \sim Z$	$\frac{\bar{x} - \bar{y} - (\mu_X - \mu_Y)}{\sigma} \sim Z$	?	$\frac{\bar{x} - \bar{y} - (\mu_X - \mu_Y)}{\sigma} \sim Z$	$\frac{\bar{x} - \bar{y} - (\mu_X - \mu_Y)}{\sigma} \sim Z$
σ: not known	$\frac{\bar{x} - \bar{y} - (\mu_X - \mu_Y)}{s} \sim T_{n-1}$	$\frac{\bar{x} - \bar{y} - (\mu_X - \mu_Y)}{s} \sim Z$	?	$\frac{\bar{x} - \bar{y} - (\mu_X - \mu_Y)}{s} \sim T_{n-1}$? ?	$\frac{\bar{x} - \bar{y} - (\mu_X - \mu_Y)}{s} \sim Z$

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Hypothesis/Significance Tests

HYPOTHESIS	SIGNIFICANCE
Parameter of interest and its estimator	Parameter of interest and its estimator
Null/alternative hypotheses	Null/alternative hypotheses
Test statistic and its distribution (assuming the null hypothesis is true)	Test statistic and its distribution (assuming the null hypothesis is true)
Rejection region	Observe the sample and calculate the test statistics or the estimator
Observe the sample and calculate the test statistics or the estimator	Calculate the corresponding p-value
Decide	Decide

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Hypothesis & significance testing

Hypothesis testing:

$\alpha \rightarrow \underbrace{\text{Rejection Region}}_{\text{computation of the critical value, k}} \rightarrow \underbrace{\text{conclusion}}_{\text{comparison of } (\bar{x} \text{ and k}) \text{ or } (z \text{ and } z_\alpha)}$

$\bar{x}$ is in the rejection region → reject H_0 at level α
$\bar{x}$ is not in the rejection region → cannot reject H_0 at level α

Significance testing:

$\bar{x} \rightarrow \underbrace{P\{\text{type I error when } H_0 \text{ is rejected with } k = \bar{x}\}}_{\text{computation of the P-value, P}} \rightarrow \underbrace{\text{conclusion}}_{\text{comparison of P-value and } \alpha}$

P-value ≤ α → reject H_0 at level α
P-value > α → cannot reject H_0 at level α

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Large-sample tests for a population proportion

X : # of successes in a random sample with n << population size

p : proportion of 'success' in a population

X~bin(n,p)

- For large n with np ≥ 10 and n(1-p) ≥ 10, X is approximately normal with $E(X) = np$, & $V(X) = np(1-p)$.
- $\hat{p} = X/n$ is unbiased since $E(\hat{p}) = p$, has approximately a normal distribution, and its standard deviation is $\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$.

H_0: $p = p_0$

true only if $np_0 \geq 10$ & $n(1-p_0) \geq 10$

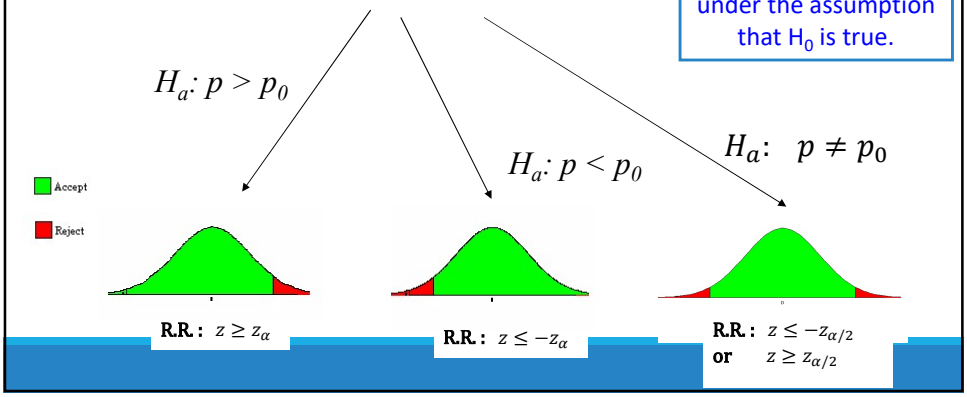
When H_0 is true, $Z = \dfrac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}} \sim N(0,1)$

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Test statistics, H_a & RR for population proportion p

$H_0: p = p_0$ **Test statistics:** $Z = \dfrac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}}$

Test statistics is given under the assumption that H_0 is true.

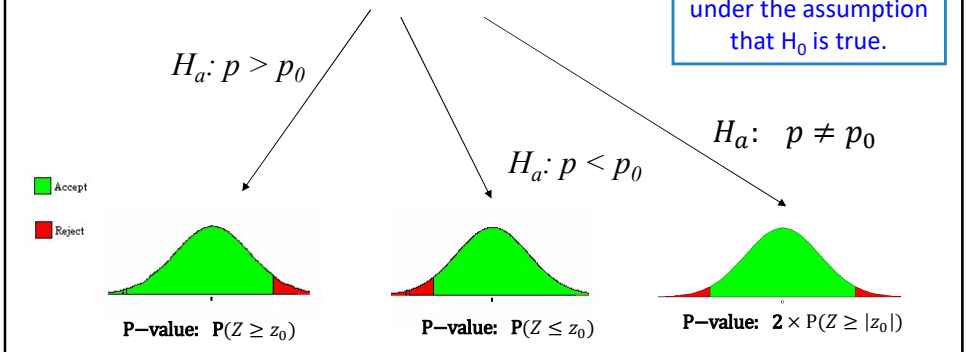


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Test statistics, H_a & p-value for population proportion p

$H_0: p = p_0$ **Test statistics:** $z_0 = \dfrac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}}$

Test statistics is given under the assumption that H_0 is true.



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Class exercise

A grinding machine will be qualified for a particular task if it can be shown to produce less than 8% defective parts. In a random sample of 300 parts, 12 were defective. Based on this data, can the machine be qualified? To answer this question:

a) Formulate the hypothesis test.

b) Find the p-value.

c) What is your conclusion for α=1%? How about α=5%?

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Hypothesis Testing for two populations

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Tests about means of two normal populations with known variance

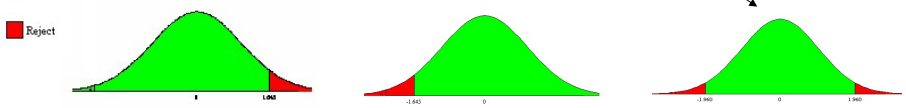
H_0: $\mu_1 - \mu_2 = \Delta_0$

Test statistics: $Z = \dfrac{\bar{X} - \bar{Y} - \Delta_0}{\sqrt{\dfrac{\sigma_1^2}{m} + \dfrac{\sigma_2^2}{n}}}$

H_a: $\mu_1 - \mu_2 > \Delta_0$

H_a: $\mu_1 - \mu_2 \neq \Delta_0$

H_a: $\mu_1 - \mu_2 < \Delta_0$



R.R.: $z \geq z_\alpha$

R.R.: $z \leq -z_\alpha$

R.R.: $z \leq -z_{\alpha/2}$ or $z \geq z_{\alpha/2}$

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Example

The production manager wants to compare the average weights of popcorns from two production lines. Experience shows that the weights of popcorns can be assumed to be normal. He would like to know whether the true mean of the weights of popcorns from two production lines are different than each other. He obtained the following data:

	sample size	sample mean	σ pop. std. deviation
Line 1	12	5.2	0.5
Line 2	16	5.6	0.6

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Example

	sample size	sample mean	σ pop. std. deviation
Line 1	12	5.2	0.5
Line 2	16	5.6	0.6

The production manager wants to compare the average weights of popcorns from two production lines. Experience shows that the weights of popcorns can be assumed to be normal. He would like to know whether the true mean of the weights of popcorns from two production lines are different than each other.

Formulate the appropriate hypothesis testing.

$$H_0: \ \mu_1 = \mu_2$$
$$H_a: \ \mu_1 \neq \mu_2$$

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Example

	sample size	sample mean	σ pop. std. deviation
Line 1	12	5.2	0.5
Line 2	16	5.6	0.6

The production manager wants to compare the average weights of popcorns from two production lines. Experience shows that the weights of popcorns can be assumed to be normal. He would like to know whether the true mean of the weights of popcorns from two production lines are different than each other.

$$H_0: \ \mu_1 = \mu_2$$
$$H_a: \ \mu_1 \neq \mu_2$$

Let α=0.05. Can you reject H_0?

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Example

	sample size	sample mean	σ pop. std. deviation
Line 1	12	5.2	0.5
Line 2	16	5.6	0.6

The production manager wants to compare the average weights of popcorns from two production lines. Experience shows that the weights of popcorns can be assumed to be normal. He would like to know whether the true mean of the weights of popcorns from two production lines are different than each other.

$$H_0: \ \mu_1 = \mu_2$$
$$H_a: \ \mu_1 \neq \mu_2$$

What is the p-value of the test for the given data?

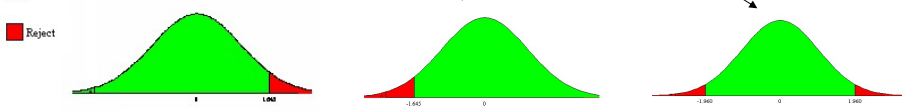
Tests about means of two populations with large sample size

$H_0: \ \mu_1 - \mu_2 = \Delta_0$ **Test statistics:** $Z = \dfrac{\bar{X} - \bar{Y} - \Delta_0}{\sqrt{\dfrac{S_1^2}{m} + \dfrac{S_2^2}{n}}}$

$H_a: \ \mu_1 - \mu_2 > \Delta_0$

$H_a: \ \mu_1 - \mu_2 < \Delta_0$

$H_a: \ \mu_1 - \mu_2 \neq \Delta_0$



R.R.: $z \geq z_\alpha$

R.R.: $z \leq -z_\alpha$

R.R.: $z \leq -z_{\alpha/2}$ **or** $z \geq z_{\alpha/2}$

Tests about means of two normal populations with unknown variance

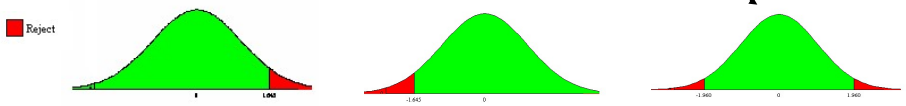
H_0: $\mu_1 - \mu_2 = \Delta_0$

Test statistics: $t = \dfrac{\bar{x} - \bar{y} - \Delta_0}{\sqrt{\dfrac{s_1^2}{m} + \dfrac{s_2^2}{n}}}$

H_a: $\mu_1 - \mu_2 > \Delta_0$

H_a: $\mu_1 - \mu_2 < \Delta_0$

H_a: $\mu_1 - \mu_2 \neq \Delta_0$



R.R.: $t \geq t_{\alpha,\nu}$

R.R.: $t \leq -t_{\alpha,\nu}$

R.R.: $t \leq -t_{\alpha/2,\nu}$ **or** $t \geq t_{\alpha/2,\nu}$

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Example: If the variances are unknown

	sample size	sample mean	Sample std. deviation
Line 1	12	5.2	0.5
Line 2	16	5.6	0.6

The production manager wants to compare the average weights of popcorns from two production lines. Experience shows that the weights of popcorns can be assumed to be normal. He would like to know whether the true mean of the weights of popcorns from two production lines are different than each other.

H_0: $\mu_1 = \mu_2$
H_a: $\mu_1 \neq \mu_2$

Sample standard deviations are: $s_1 = 0.5$ and $s_2 = 0.6$. Note that the corresponding degrees of freedom is 25.

Let α=0.05. Can you reject H_0?

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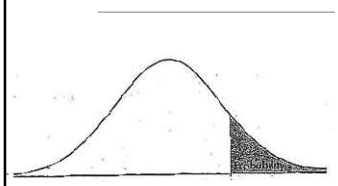


TABLE B: *t*-DISTRIBUTION CRITICAL VALUES

	Tail probability p											
df	.25	.20	.15	.10	.05	.025	.02	.01	.005	.0025	.001	.0005
1	1.000	1.376	1.963	3.078	6.314	12.71	15.89	31.82	63.66	127.3	318.3	636.6
2	.816	1.061	1.386	1.886	2.920	4.303	4.849	6.965	9.925	14.09	22.33	31.60
3	.765	.978	1.250	1.638	2.353	3.182	3.482	4.541	5.841	7.453	10.21	12.92
4	.741	.941	1.190	1.533	2.132	2.776	2.999	3.747	4.604	5.598	7.173	8.610
5	.727	.920	1.156	1.476	2.015	2.571	2.757	3.365	4.032	4.773	5.893	6.869
6	.718	.906	1.134	1.440	1.943	2.447	2.612	3.143	3.707	4.317	5.208	5.959
7	.711	.896	1.119	1.415	1.895	2.365	2.517	2.998	3.499	4.029	4.785	5.408
8	.706	.889	1.108	1.397	1.860	2.306	2.449	2.896	3.355	3.833	4.501	5.041
9	.703	.883	1.100	1.383	1.833	2.262	2.398	2.821	3.250	3.690	4.297	4.781
10	.700	.879	1.093	1.372	1.812	2.228	2.359	2.764	3.169	3.581	4.144	4.587
11	.697	.876	1.088	1.363	1.796	2.201	2.328	2.718	3.106	3.497	4.025	4.437
12	.695	.873	1.083	1.356	1.782	2.179	2.303	2.681	3.055	3.428	3.930	4.318
13	.694	.870	1.079	1.350	1.771	2.160	2.282	2.650	3.012	3.372	3.852	4.221
14	.692	.868	1.076	1.345	1.761	2.145	2.264	2.624	2.977	3.326	3.787	4.140
15	.691	.866	1.074	1.341	1.753	2.131	2.249	2.602	2.947	3.286	3.733	4.073
16	.690	.865	1.071	1.337	1.746	2.120	2.235	2.583	2.921	3.252	3.686	4.015
17	.689	.863	1.069	1.333	1.740	2.110	2.224	2.567	2.898	3.222	3.646	3.965
18	.688	.862	1.067	1.330	1.734	2.101	2.214	2.552	2.878	3.197	3.611	3.922
19	.688	.861	1.066	1.328	1.729	2.093	2.205	2.539	2.861	3.174	3.579	3.883
20	.687	.860	1.064	1.325	1.725	2.086	2.197	2.528	2.845	3.153	3.552	3.850
21	.686	.859	1.063	1.323	1.721	2.080	2.189	2.518	2.831	3.135	3.527	3.819
22	.686	.858	1.061	1.321	1.717	2.074	2.183	2.508	2.819	3.119	3.505	3.792
23	.685	.858	1.060	1.319	1.714	2.069	2.177	2.500	2.807	3.104	3.485	3.768
24	.685	.857	1.059	1.318	1.711	2.064	2.172	2.492	2.797	3.091	3.467	3.745
25	.684	.856	1.058	1.316	1.708	2.060	2.167	2.485	2.787	3.078	3.450	3.725
26	.684	.856	1.058	1.315	1.706	2.056	2.162	2.479	2.779	3.067	3.435	3.707
27	.684	.855	1.057	1.314	1.703	2.052	2.158	2.473	2.771	3.057	3.421	3.690
28	.683	.855	1.056	1.313	1.701	2.048	2.154	2.467	2.763	3.047	3.408	3.674
29	.683	.854	1.055	1.311	1.699	2.045	2.150	2.462	2.756	3.038	3.396	3.659
30	.683	.854	1.055	1.310	1.697	2.042	2.147	2.457	2.750	3.030	3.385	3.646
40	.681	.851	1.050	1.303	1.684	2.021	2.123	2.423	2.704	2.971	3.307	3.551
50	.679	.849	1.047	1.299	1.676	2.009	2.109	2.403	2.678	2.937	3.261	3.496
60	.679	.848	1.045	1.296	1.671	2.000	2.099	2.390	2.660	2.915	3.232	3.460
80	.678	.846	1.043	1.292	1.664	1.990	2.088	2.374	2.639	2.887	3.195	3.416
100	.677	.845	1.042	1.290	1.660	1.984	2.081	2.364	2.626	2.871	3.174	3.390
1000	.675	.842	1.037	1.282	1.646	1.962	2.056	2.330	2.581	2.813	3.098	3.300
∞	.674	.841	1.036	1.282	1.645	1.960	2.054	2.326	2.576	2.807	3.091	3.291

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Example: Unknown variances

	sample size	sample mean	Sample std. deviation
Line 1	12	5.2	0.5
Line 2	16	5.6	0.6

The production manager wants to compare the average weights of popcorns from two production lines. Experience shows that the weights of popcorns can be assumed to be normal. He would like to know whether the true mean of the weights of popcorns from two production lines are different than each other.

$$H_0: \ \mu_1 = \mu_2$$
$$H_a: \ \mu_1 \neq \mu_2$$

Sample standard deviations are: $s_1 = 0.5$ and $s_2 = 0.6$. Note that the corresponding degrees of freedom is 25.

What is the p-value of the test for the given data? What can you say about the p-value of the test for the given data?

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Example: back to known variance & type-2 error

	sample size	sample mean	σ pop. std. deviation
Line 1	12	5.2	0.5
Line 2	16	5.6	0.6

The production manager wants to compare the average weights of popcorns from two production lines. Experience shows that the weights of popcorns can be assumed to be normal. He would like to know whether the true mean of the weights of popcorns from two production lines are different than each other.

$$H_0: \ \mu_1 = \mu_2$$
$$H_a: \ \mu_1 \neq \mu_2$$

Find the type-II error, β, when $\mu_1 - \mu_2 = -0.3$ with $\alpha = 0.05$?

Inferences concerning a difference between population proportions

$X \sim Bin(m, p_1) \quad Y \sim Bin(n, p_2) -$ independent

Estimators for proportions: $\hat{p}_1 = \frac{X}{m}$ and $\hat{p}_2 = \frac{Y}{n}$

$$E\left[\frac{X}{m} - \frac{Y}{n}\right] = p_1 - p_2 \qquad V(\hat{p}_1 - \hat{p}_2) = \frac{p_1 q_1}{m} + \frac{p_2 q_2}{n}$$

large $m\&n$: $$Z = \frac{\hat{p}_1 - \hat{p}_2 - (p_1 - p_2)}{\sqrt{\frac{p_1 q_1}{m} + \frac{p_2 q_2}{n}}} \sim N(0,1)$$

Tests concerning a difference between population proportions

$H_0: p_1 - p_2 = \Delta_0$

Test statistics: $z_0 = \dfrac{\hat{p}_1 - \hat{p}_2 - \Delta_0}{\sqrt{\dfrac{\hat{p}_1\hat{q}_1}{m} + \dfrac{\hat{p}_2\hat{q}_2}{n}}}$

$H_a: p_1 - p_2 > \Delta_0$

$H_a: p_1 - p_2 \neq \Delta_0$

$H_a: p_1 - p_2 < \Delta_0$



P−value: $P(Z \geq z_0)$

P−value: $P(Z \leq z_0)$

P−value: $2 \times P(Z \geq |z_0|)$

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Most Common H_0: p_1=p_2

- Estimators for proportions: $\hat{p}_1 = \dfrac{X}{m}$ and $\hat{p}_2 = \dfrac{Y}{n}$
- If H_0 is true, $p = p_1 = p_2$, then p can be estimated by $\hat{p}$.

$$\hat{p} = \frac{m}{m+n}\hat{p}_1 + \frac{n}{m+n}\hat{p}_2 = \frac{X+Y}{m+n}$$

- Then, we change our estimator for the standard deviation:

$$Z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}\hat{q}\left(\dfrac{1}{m} + \dfrac{1}{n}\right)}} \sim N(0,1)$$

z	0.00	0.01	0.02	
0.0	0.5000	0.5040	0.5080	0.
0.1	0.5398	0.5438	0.5478	0.
0.2	0.5793	0.5832	0.5871	0.
0.3	0.6179	0.6217	0.6255	0.
0.4	0.6554	0.6591	0.6628	0.
0.5	0.6915	0.6950	0.6985	0.
0.6	0.7257	0.7291	0.7324	0.
0.7	0.7580	0.7611	0.7642	0.
0.8	0.7881	0.7910	0.7939	0.
0.9	0.8159	0.8186	0.8212	0.
1.0	0.8413	0.8438	0.8461	0.
1.1	0.8643	0.8665	0.8686	0.

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Class exercise

The article "Factors Associated with Exercise Behavior in People with Parkinson Disease" (T. Ellis, J. Cavanaugh, et al., *Physical Therapy*, 2011:1838–1848) reported a survey of patients with Parkinson's disease. Of 164 patients who said they exercised regularly, 76 reported falling in the previous six months. Of 96 patients who said they did not exercise regularly, 48 reported falling in the previous six months. Can you conclude that the proportion of patients who fall is less for those who exercise than for those who do not?

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

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Example

- Depression treatment
- A certain extract versus placebo
- 200 patients: randomly assigned to one of the treatments
- Placebo: 19 patients improved out of 100
- Extract: 27 patients improved out of 100
- State the appropriate hypotheses.
- Find the corresponding p-value.
- What is your conclusion?

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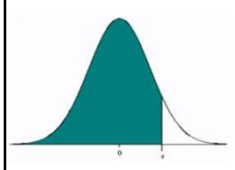


Table of Standard Normal Probabilities for Positive Z-scores

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

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